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Procedia Computer Science 258 (2025) 2119–2129

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Machine Learning and Data Engineering

Enhancing Traffic Monitoring Systems through Image Analysis and Rule-Based Machine Learning

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Abstract

This paper presents an advanced traffic monitoring system integrating reasoning-based image analysis and rule-based machine learning. Reasoning-based image analysis interprets visual data to extract meaningful information, identify objects, recognize patterns, and make logical deductions. Our system leverages visual data from spot sensors to detect traffic objects using a three-tier module approach, enhancing detection accuracy across various illumination levels throughout the day. At the higher level, rule-based reasoning refines the model, complementing artificial intelligence with low-level image analysis for increased flexibility and real-time responsiveness. By processing visual data from traffic cameras, our system identifies vehicles, tracks movements, and detects violations. The rule-based machine learning models predict traffic patterns and identify anomalies, trained on extensive datasets for improved accuracy and adaptability. Evaluated through simulations and real-world testing, this integrated approach significantly advances intelligent transportation systems by improving traffic management, reducing congestion, and enhancing road safety.

Keywords: Rule based reasoning; Image analysis; RCNN; Cloud processing; Traffic monitoring

1. Introduction

The world in this phase of technological advancements embraces ideas and innovative solutions to address day-to- day challenges. AI in the transportation industry is reshaping the entire world. Be it optimization of traffic management or multiple applications of AI in transportation, there is a clear exponential surge in research and development for

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10.1016/j.procs.2025.04.462

adopting the technology. Using machine learning for traffic analysis involves combining image analysis and rule-based reasoning to interpret traffic patterns, make predictions, and suggest optimizations. Traffic data may be captured by a variety of sensors nowadays, including high-end cameras, unlike earlier traditional methods involving spot sensors like pneumatic sensors and loop detectors. The high resolution vision based cameras are capable of capturing richer data dynamically, adapting to diverse conditions, and addressing the complexities of urban traffic management more effectively than traditional spot sensors. These advanced systems leverage image sequences to provide detailed information for precise vehicle tracking and classification, surpassing the capabilities of spot sensors limited to basic measurements like traffic flow. They adapt to various conditions and provide a comprehensive view of traffic dynamics, adjusting easily to different environments, weather conditions, and lighting scenarios.

Urban environments pose unique challenges due to their highly variable background conditions. Vision-based systems excel in such complex scenarios by processing dynamic visual data to understand and respond to traffic patterns. The approach we propose utilizes this multi-condition adaptability feature of vision-based cameras. While placing a camera helps to see, what is done after seeing is crucial. In this approach we adopt a three level model: the high-level reasoning module, the middle-level interface module, and a specialized low-level image-processing module. This system integrates reasoning-based image analysis and rule-based machine learning to enhance traffic monitoring. The focus of the paper is to perform round-the-clock traffic monitoring with the least computational load to work efficiently in any condition. Two different modules separated based on the intensity of light i.e. Morphological analysis at night on headlight pairs and Google cloud processing using RCNN algorithm in daytime make the paper ideal to suit the requirements of the real-time traffic scenarios. Study efforts elaborated in the literature section also highlight the different ways to capture, process, and track vehicles using spot sensors, image processing, and analysis.

The latter content of the paper is organized with an elaborate literature review of the different methodologies and studies to enhance traffic monitoring systems. The following section “Materials and Methods” illustrates the three different modules further explained in the subsections followed by the experimental results and evaluation metrics. The conclusion section gives a brief on the challenges faced, restrictions or checks, and future scope.

2. Literature Review

In the present scenario, numerous studies have explored different methodologies to enhance traffic monitoring systems through advanced image analysis and rule-based machine learning techniques. These studies aim to improve accuracy, real-time compatibility, and adaptability to varying conditions, focusing on both daytime and nighttime traffic monitoring. M. Taha (2014) proposed a method using automatic thresholding combined with a rule-based analysis component [1], achieving an impressive accuracy of 96.27% on urban roads. This approach demonstrated high effectiveness in detecting traffic objects. However, it highlighted a need for further research to improve the detection of low-visibility vehicles, a common challenge during adverse weather conditions or nighttime.

H. W. Huang (2017) developed a traffic monitoring system based on spatiotemporal analysis [2] using RCC sensors. This method attained an accuracy of 92.55%, providing stable tracking capabilities. Despite its effectiveness, the system's performance was significantly hindered by poor image quality, which could be attributed to various environmental factors such as low light or occlusions. Piotr B. (2019) introduced a technique that combined Neural-Networks in its CNN form with HOG (Histogram – of – Oriented - Gradients) features [3]. The following approach reached an accuracy level of 93.63%, benefiting from the powerful feature extraction capabilities of CNNs. However, the longer training period required for the neural network was a notable drawback, making it less practical for real-time applications.

Y. L. Murphey (2019) proposed an intelligent system for moving target detection utilizing fuzzy rules [4]. While the exact accuracy was not specified, this method was effective in detecting moving targets through the application of fuzzy rules. The limitation, however, was that it primarily focused on non-symbolic visuals, which restricted its application to broader traffic monitoring needs. Chen J. C. (2019) applied Region-based Convolutional Neural Networks (RCNN) [5] for vehicle detection during both day and night. Achieving an accuracy of 95.1%, this system proved to be highly effective in varying lighting conditions. Nonetheless, the performance was compromised when there were changes in daylight intensity due to bad weather, which could affect the reliability of the detection system. Ligayo M. A. D. (2021) implemented YOLOv3 and CNN for detecting vehicle headlights [6] during both day and night, achieving an accuracy of 86.72%. This approach was particularly effective in detecting headlights, demonstrating robustness across different lighting conditions. However, it did not incorporate the detection of other vehicles captured by the camera, limiting its overall effectiveness in comprehensive traffic monitoring.

These studies collectively highlight significant advancements in traffic monitoring through the integration of machine learning and image analysis techniques. The use of high-resolution cameras and advanced algorithms has improved the accuracy and reliability of traffic object detection. However, challenges remain in terms of real-time adaptability, handling low-visibility conditions, and ensuring consistent performance across diverse environments. Our proposed system seeks to address these challenges by leveraging a three-level module approach that combines high-level reasoning, a middle-level interface, and specialized low-level image processing. This comprehensive system aims to enhance the accuracy of traffic monitoring across different illumination levels and environmental conditions. By integrating vision-based high-resolution camera systems, our approach provides richer data and more precise vehicle tracking and classification. The rule-based reasoning module at the higher level further refines the model, enabling better real-time responsiveness and flexibility.

Table 1. Literature review

Author	Year	Technique	Accuracy	Advantages	Disadvantages
M. Taha	2014	Automatic thresholding with Rule-based analysis	96.27%	High accuracy on urban roads	Low-visibility vehicles need further research
H. W. Huang	2017	Spatiotemporal analysis using RCC sensors	92.55%	Stable tracking	Poor image quality reduces effectiveness
Piotr B.	2019	Convolutional Neural Networks (CNN) and HOG features	93.63%	High accuracy	Longer training period
Y. L. Murphey	2019	Intelligent system for moving target detection using fuzzy rules	Not specified	Effective moving target detection with fuzzy rules	Applied only to non- symbolic visuals
Chen J. C.	2019	RCNN for vehicle detection day and night	95.1%	High accuracy both during day and night	Compromised performance with daylight intensity changes
Ligayo M. A. D.	2021	YOLOv3, CNN for detecting headlights day and night	86.72%	Effective headlight detection day and night	Did not incorporate detection of other vehicles

A summary for the different techniques used in literature survey has been listed in Figure 1.

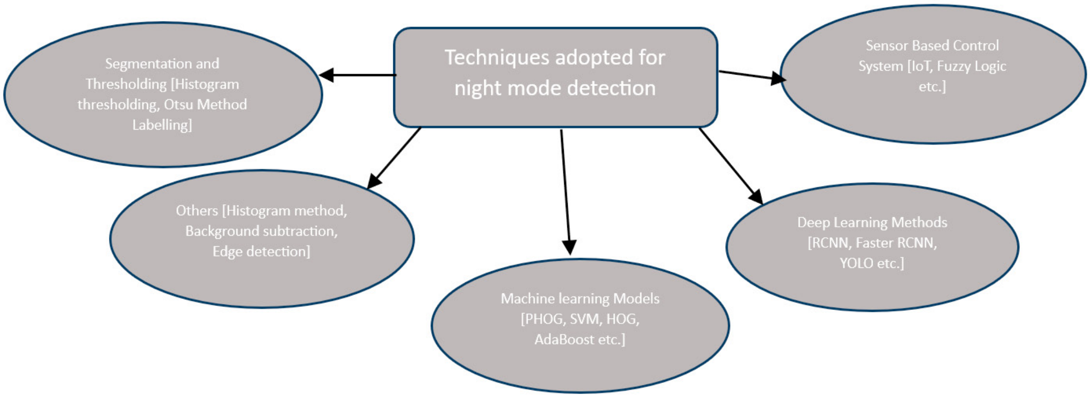

Fig. 1. Different techniques used in literature survey.

3. Materials and Methods

The proposed traffic monitoring system leverages a three-level modular approach to improve detection accuracy under varying illumination conditions. The lowest-level module captures image streams, distinguishing between low and good light intensities, which are then processed separately. Nighttime object extraction focuses on headlight detection using a bimodal histogram and Otsu's thresholding, ensuring effective separation of foreground and background. Daytime processing involves masking unwanted objects and detecting moving points using a double-difference operation for accurate target extraction. The intermediate module preprocesses data using HSV color masking and spatiotemporal segmentation, reducing complexity and filtering noise. The processed output is transferred to the cloud for final analysis using RCNN, enabling robust vehicle detection and classification. The higher third level module incorporates a knowledge oriented expert model with rules based reasoning, combining domain-specific expertise and real-time data to adapt to dynamic traffic scenarios. This module includes a knowledge base, inference engine, and context memory, ensuring continuous decision-making. The interface system facilitates knowledge acquisition and provides explanations, enhancing user interaction. This comprehensive approach integrates visual and symbolic data, offering improved traffic monitoring and management capabilities.

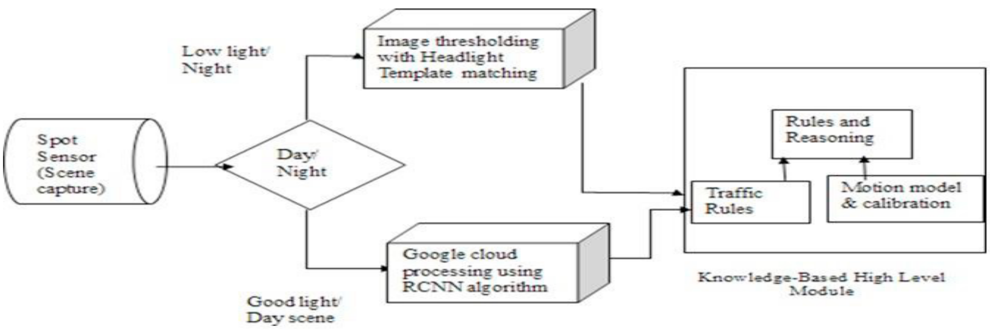

Fig. 2. Detailed steps.

The processing steps primarily involve the lowest-level module capturing the image stream directly from a traffic scene. At this stage, the system categorizes the light intensity as either low or good and channels the data into two distinct paths. Each channel processes scenes according to their light intensity. As shown in Figure 2, the processed outputs from both channels are then fed into the rule-based expert system. This expert system utilizes rule-based reasoning, applying common sense and domain-specific expertise in the form of conditional rules [i.e., IF<preconditions> THEN <conclusions>]. This method ensures accurate interpretation and response to varying traffic conditions, enhancing the overall traffic monitoring system's effectiveness.

3.1. Module 1

Object extraction at night differs significantly from daytime due to reduced illumination as the day progresses. During the night, headlight beams become the primary visual features targeted for detection, as illustrated in Figure 2. Spot sensor-like cameras, strategically placed, can detect headlight beams over a large area. To simplify the image, distracting objects such as pavements, trees, and street lamps are masked out. The system then extracts target objects using a bimodal histogram from the background.

At night, images can be easily binarized due to the generally low grey level. Binarization involves separating the complete image into foreground class and background class based on the grayscale intensity values of its pixels, employing Otsu's method. This method calculates a threshold that minimizes the intra-class variance, effectively distinguishing headlights from the darker background. The lighter regions, representing the headlights, are isolated for further processing, while the darker background is discarded. This approach ensures accurate detection of vehicles by focusing on their headlights, enhancing the system's capability to monitor traffic efficiently under low-light conditions.

Good light conditions are not available at night for traffic data extraction [7]. So, the spot sensor-like camera is utilized to detect headlight beams over a large area based on its placement [8] in a low light-conditioned environment. The headlight beams are the target visual features Fig. 3. to be extracted while distracting objects are masked out to make the image simpler (i.e. pavements, trees, street lamps, etc.) followed by extraction of the target objects implementing a bimodal histogram [9] from the background. Easily the image is binarized due to low grey level in the night. Gaussian blur is used to reduce noise. Compute the histogram of the grayscale image. Finally, separate the complete image class into two (foreground and background) employing Otsu's method. The steps involved are Preprocessing, Segmentation, and feature extraction.

3.1.1. Preprocessing

The preprocessing involves converting the image to grayscale followed by Noise reduction [10] using Gaussian blur denoising technique. Then compute the histogram of the grayscale image for further processing.

3.1.2. Segmentation and feature extraction

The optimal threshold that separates the two peaks in the bimodal histogram is determined by the Otsu Method [11] due to its simplicity and effectiveness. This technique can automatically determine the threshold(optimal) by maximizing or minimizing intraclass variance for segmentation by assuming that the image class comprises of two categories of pixels for the background as well as the foreground. The lighter region with a darker background square template is used for the headlight template match. Finally, the headlight pairs are traced or extracted for feature count by cross-correlation using scaled correlation parameters.

{Grayscale histogram H(I), M is the width and N the height of I(Image) with intensity i}

$$H(I) \sum_{x=1}^{M} \sum_{y=1}^{N} [I(x, y) = i]$$

{Cumulative Distribution Function C(i)}

$$C(i) \sum_{j=0}^{i} H(j)/M*N \tag{1}$$

{Mean Grayscale Intensity μ}

$$\mu = \sum_{x=1}^{M} \sum_{y=1}^{N} (x, y)/(M * N) \tag{2}$$

{Class variance - difference of the foreground region and background region.}

$$var(T) = P0\,(T). P1\,(T). (m0\,(T) - m1\,(T))^2 \tag{3}$$

{probabilities of the background and foreground}

$$Po(T) = C(T) \tag{4}$$

{probabilities of the background and foreground}

$$P1\,(T) = 1 - C(T) \tag{5}$$

{mean grayscale intensity values of the background}

$$m0\,(T) = \sum_{i=0}^{T-1} i \,. \frac{H(i)}{[(P0\,(T).MN)]} \tag{6}$$

{mean grayscale intensity values of the foreground regions}

$$m1\,(T) = \sum_{i=T}^{T-1} i \,. \frac{H(i)}{[(P1\,(T).MN)]} \tag{7}$$

{Optimal threshold}

$$Topt = argmaxT(var(T)) \tag{8}$$

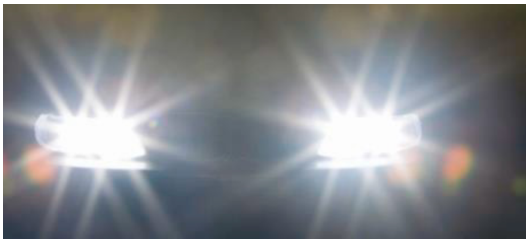

Fig. 3. Captured source

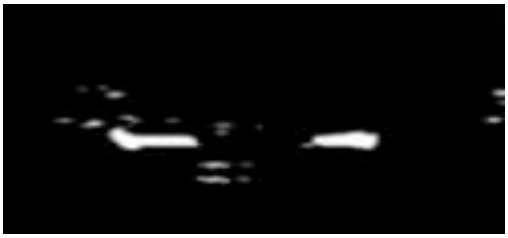

Fig. 4. Grey scale conversion of captured source

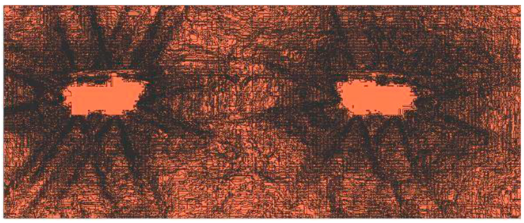

Fig. 5. Noise reduction.

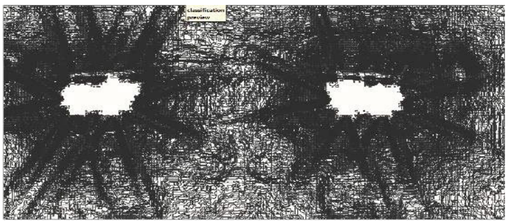

Fig. 6. Grey scale conversion

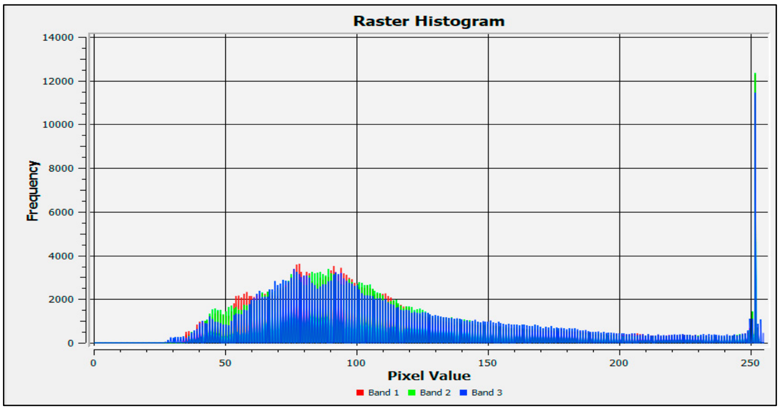

Fig. 7. Histogram computation

3.2. Module 2

In the daytime, the scene captured comprises of traffic (TARGET) and other objects (Unwanted) such as pavement, street lights, houses, etc. Extracting target data in the daytime is easier as light conditions are good. During the daytime, the use of headlights is absent hence a different flowchart of methods is required. Figure. 8. Explains the complete workflow for module 2.

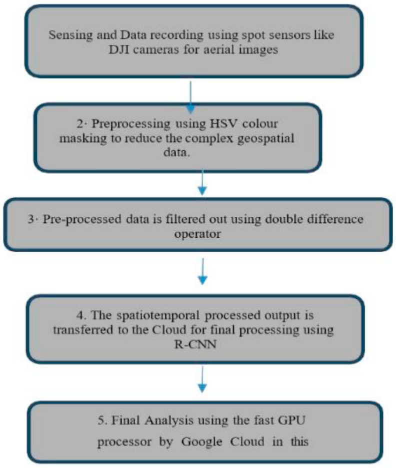

Fig. 8. Flow diagram representing the processing in Module2

Pre-processing the captured source involves the masking of unwanted objects and greenery areas. The collected dataset is cropped and resized to reduce size like eliminating roadside pavements etc. HSV is applied to remove the Greenery aspect from the scene available to reduce the dataset further. Though this masking separates a major percentage of unwanted features from the target, it cannot completely extract the target object from the complete scene.

Hence the preprocessed imagery is further processed. Here extraction is based on the detection of moving objects i.e. moving points by double-difference operation (i.e. 3 frame difference) [12-13].

$$D_n(i,j) = |I_n(i,j) - I_{n-1}(i,j)| \{\text{Difference image } D_n \text{ and Binary Image is } I_n\} \quad (9)$$

$$DD_n(i,j) = \{1 \; or \; 0\} \; \{\text{Double Difference image}\} \quad (10)$$

$$MEC_n^0(i,j) = \{1 \; if \; DD_n(i,j) = 1 \; or \; 0\} \; \{\text{moving-edge closure}\} \quad (11)$$

$$MEC_n^r(\mathrm{i,j}) = \{1 \text{ or } 0\}$$
$$\{ \, if \; MEC_n^{r-1}(i,j) = 1 \vee (\nabla In(i,j) > T_G \wedge \ni (k,l) \epsilon X | MEC_n^{r-1}(k,l) = 1 \; \; or \; Otherwise \, \} \quad (12)$$

Studies prove that this operation has high accuracy and can locate even tiny objects. It also easily filters out noise resulting from camera movements. This is a spatiotemporal segmentation process as it processes images of variant luminance(spatial) and extracts temporary existing points in a frame (temporal) w.r.t time. The spatiotemporal processed output is transferred to the google Cloud for final processing using R-CNN.

3.3. Knowledge-Based High-Level Module 3

A missing program element in traditional programming makes the process stop. This is highly undesired for dynamic traffic scenario which demands expert judgement along with reasoning strategies to keep deciding results. The knowledge-based expert systems [14] unlike traditional programs (using algorithmic procedures) do not follow step-by-step procedures. The proposed model with a knowledge-based expert system has program-solving capability with human experts reasoning. This high-level module unlike spot sensors uses visual symbolic data to assess the present traffic situation. This can be closely illustrated with the example of two very closely moving forms travelling with the same speed may be taken to be a single object because of their close proximity due to over segmentation because of occlusions(shadows). This allows forward chaining reasoning where the current data with respect to time is in working memory and forward-inferences chain is used to reach the goal.

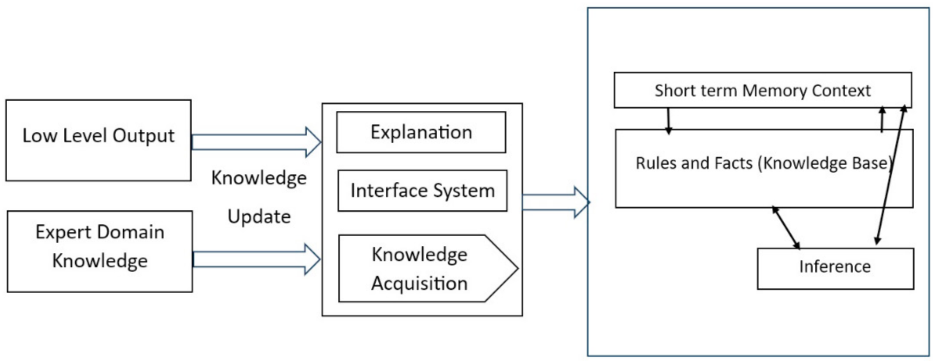

Fig. 9. Knowledge-Based High-Level Module

The components as shown in Fig 9. Comprises of the Rule-based reasoning module (Knowledge Base, Short-term Memory Context, and Inference) and the User Interface System for input and expert advice.

Knowledge Base is where the rules and facts are stored for the domain-specific system knowledge. This is needed for the analysis of a problem and its solution. This can be regarded as the powerhouse of the system containing the facts and rules that direct its usage to solve a given traffic scenario.

The Inference Engine is the brain involved in processing data by combining information supplied with the rules in the knowledge database. It controls the fundamental processing of how to conclude.

Context is the working memory, a short-term volatile memory containing dynamic knowledge and other temporary information generated by the system.

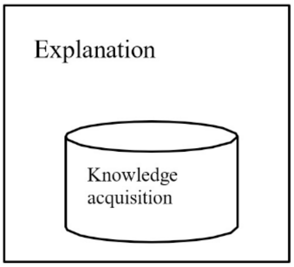

Fig. 10. Interface System.

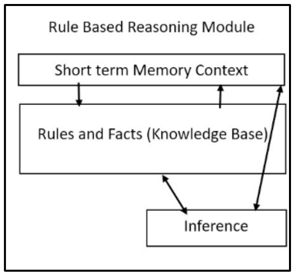

Fig. 11. Rule-Based Reasoning System.

Interface System comprises of two components. Knowledge Acquisition Module and the Explanation Module. Knowledge Acquisition Module helps in the process of organizing knowledge in the form of facts, relationships, and heuristics, from various sources (human experts, sensors, e-books) using various methods like artificial neural networks, fuzzy logic, rule-based systems, and decision trees. This is then encoded into a knowledge base for complex problem-solving and reasoning.

The Explanation Module provides reasoning, definitions, and other detailed information to the user whereas the user-friendly module for input and access is called the user interface.

4. Experimental Results and Evaluation

This tier approach having separate daytime and nighttime modules gives a maximum accuracy of 96% with two different levels of processing: a reasoning module at a higher level and a specialized image processing lower module.

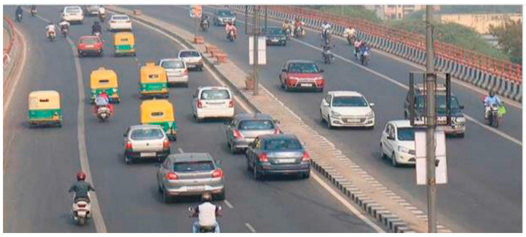

Fig. 12. Preprocessed Aerial Image (Day-Flyover)

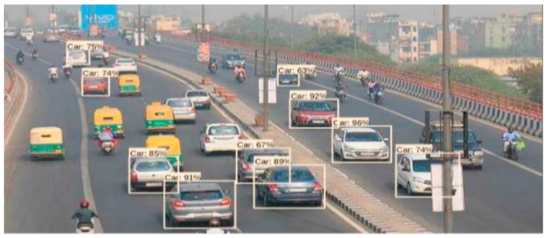

Fig. 13. Output Image(Day-Flyover)

Fig.14. Preprocessed Aerial Image(Low Night Traffic)

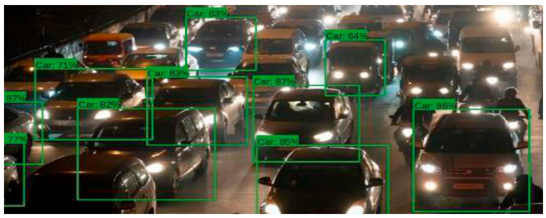

Fig. 15. Output Image(Night)

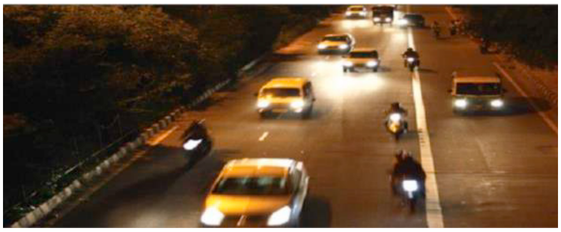

Fig.16. Preprocessed Aerial Image (Heavy Night Traffic.)

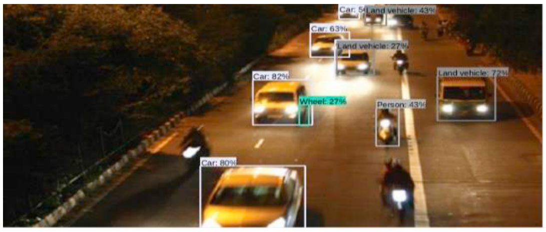

Fig. 17. Output Image (Night)

Fig. 18. Preprocessed Aerial Image (Rainy Weather.)

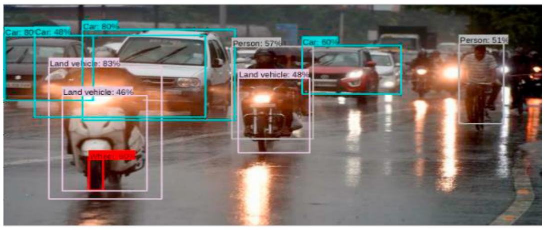

Fig. 19. Output Image (Rainy Weather.)

The robustness of a model may be evaluated from different evaluation metrics [15-16]. The robustness of this proposed three tier model in different environments and conditions has been evaluated based on the accuracy metric. The accuracy metric signifies how regularly the model predicts the output correctly. The experimental outcomes have different accuracy under different environmental conditions given in the accuracy chart of Figure 20.

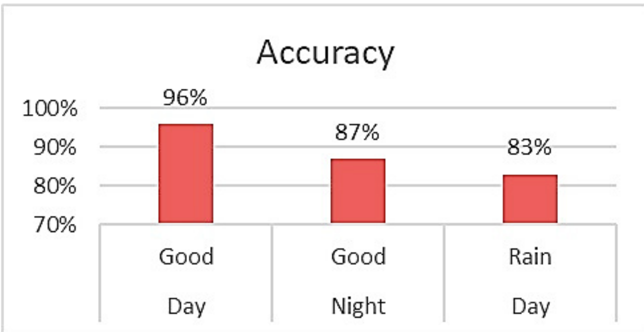

Fig. 20. Maximum accuracy

5. Conclusion

This paper puts forward the best-known methodologies adopted in the review section. Earlier various sensors were employed for the detection of vehicles, e.g. radars, ultrasound sensors, radars, etc. but they pose difficulty in real-time detection of multiple vehicles. Hence, in recent times research from the field of surveillance and intelligent vehicles has been more focused on computer-vision-based detection. However, the solutions are still insufficient to meet the needs in real-time. This work is oriented with a focus on implementing a three-tier model that proposes traffic monitoring across twenty-four hours of the day with variations in light intensity by image processing and rule-based reasoning. The framework implements three modules designed for the daytime, night vision, and reasoning modules using a forward chaining production system, exploiting a set of heuristic expert traffic rules.

The scope of real-world testing is limited to environmental conditions where different (e.g., fog, snow) conditions could not be included due to available good weather conditions. This may be taken up as future work to further validate the system's robustness.

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